

# Theory and Implementation of the Bounded Effect Size in MethylDetector

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## **Contents**

# 1 Introduction

In MethylDetector (built on MethylUtils), the **bounded effect size** is a key metric for ranking Differentially Methylated Positions (DMPs). It combines:

- **Welch’s  $d$**  (standardized mean difference, accounting for unequal variances and sample sizes).
- **Distribution overlap** (Bhattacharyya coefficient or KS statistic from ECDF).
- **Sigmoid bounding** to  $[0, 1]$  for interpretability (large effect  $\rightarrow 1$ , no effect  $\rightarrow 0$ ).

The formula is *sigmoid-like* (logistic function) and biologically interpretable: perfect separation ( $\delta\mu \gg 0$ , no overlap) yields  $\approx 1$ ; identical distributions yield  $\approx 0$ .

Two computation modes exist for efficiency:

- **Approximate** (fast, vectorized): Discrete overlap from binned histograms or Normal approximation. Used for initial screening/sorting.
- **Refined** (precise, CPU-bound): ECDF-based KS statistic via PCHIP interpolation. Used only for top positions (e.g., via MethylDetectorExplorer).

## 2 Welch’s $d$ (Standardized Mean Difference)

Welch’s  $d$  measures effect magnitude, robust to unequal variances  $n_1 \neq n_2$ :

$$d = \frac{|\mu_1 - \mu_2|}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \quad (1)$$

where:

- $\mu_1, \mu_2$ : Methylation means ( $\mu = Sx/N$  from centroid).
- $\sigma_1^2, \sigma_2^2$ : Sample variances ( $\sigma^2 = (Sx2 - Sx^2/N)/(N - 1)$ ).
- $n_1 = N_1, n_2 = N_2$ : Coverage (reads/position).

$d$  is *capped* at 50 to prevent overflow in downstream sigmoid.

## 3 Distribution Overlap Metrics

Overlap  $O \in [0, 1]$  ( $O = 1$ : identical,  $O = 0$ : no overlap) corrects  $d$  for distribution similarity.

### 3.1 Approximate Overlap ( $O_{\text{approx}}$ )

1. **Discrete Bhattacharyya** (preferred, from ‘binned\_stats’):

$$O_{\text{Bh}} = \sum_b \sqrt{p_{1b} \cdot p_{2b}}, \quad p_{ib} = \frac{\text{bin\_counts}_{ib}}{\sum_b \text{bin\_counts}_{ib}} \quad (2)$$

Vectorized over positions; fast ( $O(n_{\text{pos}} \cdot n_{\text{bins}})$ ).

2. **Normal fallback** (if no bins):

$$O_{\text{Norm}} = 2\Phi\left(-\frac{d}{2}\right) \quad (3)$$

where  $\Phi$  is standard normal CDF.

Implemented in `welch.d.fast_overlap_approx()`.

### 3.2 Refined Overlap ( $O_{\text{KS}}$ ): ECDF Kolmogorov–Smirnov

KS statistic  $D = \sup_x |F_1(x) - F_2(x)|$  from empirical CDFs ( $F_i$  via PCHIP on `bin_edges`, `bin_counts`):

$$O_{\text{KS}} = 1 - D \quad (4)$$

Implemented in `welch_d_ks_overlap()` with `ECDFView._cdf_batch()`.

## 4 Bounded Effect Size Formula

Core formula (both approx/refined):

$$\text{effect} = \sigma(s \cdot d \cdot (1 - O)) \quad (5)$$

where:

- $\sigma(z) = \frac{1}{1+e^{-z}}$  (logistic sigmoid, bounds to  $[0, 1]$ ).
- $s = 4.0$  (scale hyperparameter;  $s \cdot d_{\text{max}} \approx 200 \rightarrow \sigma \approx 1$ ).
- $O$ : Overlap ( $O_{\text{approx}}$  or  $O_{\text{KS}}$ ).

**Interpretation:**

- $d \rightarrow 0$ : effect  $\rightarrow 0$ .
- $O \rightarrow 1$ : effect  $\rightarrow 0$  (overlap dominates).
- $d \gg 0, O \rightarrow 0$ : effect  $\rightarrow 1$  (perfect discrimination).

## 5 Approximate vs Refined Computation

|                | Approximate ( $O_{\text{approx}}$ )        | Refined ( $O_{\text{KS}}$ )       |
|----------------|--------------------------------------------|-----------------------------------|
| Overlap source | Discrete bins or Normal                    | ECDF KS (PCHIP)                   |
| Complexity     | $O(n \cdot b)$ ( $b \approx 20$ )          | $O(n \cdot g)$ ( $g = 256$ grid)  |
| Use case       | Screening/sorting all positions            | Top-K refinement                  |
| Implementation | <code>welch_d_fast_overlap_approx()</code> | <code>welch_d_ks_overlap()</code> |

Table 1: Approximate vs refined bounded effect size.

## 6 MethylDetectorExplorer: Two-Phase Optimization

1. **Phase 1:** Compute  $O_{\text{approx}}$  on subsample ( $\sim 1\%$ ), sort by  $\text{effect}_{\text{approx}}$ .
2. **Phase 2:** Heuristic chooses  $K$  (decay, knee, threshold, fraction, fixed); compute  $\text{effect}_{\text{refined}}$  only for top  $K$ .

See `methyl_detector.explorer.MethylDetectorExplorer`.

## 7 Implementation Details (MethylUtils)

Files: `methyl_utils/statistical_tests.py`, `distribution_views.py`.

Key functions:

```
welch_d_fast_overlap_approx(delta_mean, var1, n1, var2, n2,
                             overlap_approx=None, scale=4.0)
    -> {"welch_d", "overlap_approx", "bounded_effect_size_approx"}

welch_d_ks_overlap(delta_mean, var1, n1, var2, n2, ecdf_view1,
                  ecdf_view2, position_indices, scale=4.0)
    -> {"welch_d", "ks_d", "ks_p", "corrected_d", "bounded_effect_size"}
```

ECDF: `ECDFView` builds per-position PCHIP interpolators from `bin_edges/bin_counts`.

## 8 References

- Welch (1947): Unequal variances t-test.
- Bhattacharyya (1943): Coefficient of distributions.
- Kolmogorov–Smirnov test.